

PROJECT PROPOSAL

Group Meeting Finder

PROBLEM

When doing group projects, groups of students can struggle to find meeting times that fit everyone's schedules and personal lives. Calendars exported as iCalendar (.ics) files contain busy blocks, but coordinating across multiple people still requires manual comparison and communication.

OBJECTIVE

Build a tool that, from several .ics calendar files and user preferences like ("no meetings after 18:00", required vs. optional attendees, duration, buffers), computes ranked candidate meeting slots and generates .ics invitations.

MODEL AND METAMODELS

Two input domains and one target domain are modeled with EMF/Ecore:

- Calendars/time - CalendarMM: Person, Calendar, Event (start/end/timezone), BusyBlock, Source.
- User preferences - PreferenceMM: ParticipantPref (work hours, no-meet windows, daily max, min buffer), MeetingRequest (duration, earliest/latest, required/optional attendees), ScoringWeights.
- Proposals/invites - ScheduleMM: CandidateSlot (interval, attendance coverage, conflicts, score, rationale), Schedule. InviteMM: abstract invitation with Invite (summary, attendees, start/end, location/online link) to be serialized to iCalendar VEVENT.

The source models CalendarMM and PreferenceMM, are independent and combined via M2M to get ScheduleMM. From ScheduleMM, M2T produces readable summaries and iCalendar invitations.

Import each member's .ics into CalendarMM (T2M) and fill PreferenceMM via for ex. CSV.

To update, re-run the M2M planner whenever inputs change,

TOOLS AND TRANSFORMATIONS

Tools. Eclipse Modeling Tools (EMF/Ecore), QVTo for Model-to-Model, and Acceleo for Model-to-Text.

Transformations to be implemented

1. T2M (Import): .ics to CalendarMM. Parse iCalendar events and time zones into EMF models.
2. M2M (Planner): (CalendarMM + PreferenceMM) to ScheduleMM. Compute free/busy timelines, generate candidate windows like 30/45/60 minutes, score by attendance coverage, preference fit, and fairness.

3. M2T (Outputs): ScheduleMM → (i) Markdown/HTML summary with a ranked table of proposals, (ii) InviteMM/.ics invitations for the selected slot(s).

COMPOSITION AND EXTERNAL TOOL INTEGRATION

Import path: T2M → M2M → M2T to produce proposals. Booking path: choose a slot then M2T to generate the final .ics. External tools produce/consume models at the boundaries, calendar apps produce .ics (consumed by T2M) and consume the generated .ics (produced by M2T).

